

# Boundary-Guided Replay: A Mechanism Study of Success–Failure Boundary Learning

Anonymous submission

## Abstract

Policies trained on nominal starts often fail under small deviations from the states encountered during training. Existing curriculum and level-replay methods prioritize difficult or high-regret levels, but they do not identify where behavior transitions from likely success to likely failure. We introduce Boundary-Guided Replay (BGR), a replay method for resettable or replayable sequential decision settings. For each replayable state, BGR estimates a recovery curve over perturbation scale, derives a critical radius, and samples training perturbations near this state-conditioned boundary. The paper is scoped as a mechanism study, not as a completed robotics or broad-benchmark result. The clearest effect is procedural grid-margin replay: BGR improves recovery AUC in both the 30-seed full-baseline run and a held-out 30-seed replication (pooled 0.4342 vs. 0.3965). Three pre-existing OpenML margin-replay checks also replicate under fixed target radii, with pooled BGR-uniform gaps from +0.0349 to +0.0858, but are supervised perturbation evidence rather than standard-environment wins. Synthetic and robot-suffix results are smaller and are reported as scoped support: the coverage-aware suffix result is manipulation-style support rather than broad robustness evidence, improving final object recovery AUC (0.4971 vs. 0.4856), clean success, transfer recovery AUC, and learning-curve area, while still trailing uniform on median critical radius. Standard-environment and OpenVLA/LIBERO audits are negative for broad robotics promotion: they measure learned-policy brittleness and build matched fine-tuning datasets, but current adaptation does not establish stable gains over the official checkpoint.

## 1 Introduction

A policy can solve nominal starts and still be brittle: a small change in layout, object pose, or trajectory prefix may move it outside its recovery basin. Standard success metrics hide this geometry. We ask a deliberately narrower question: for each replayable state, how far can the current policy be perturbed before success becomes failure, and can replay near that boundary expand the local margin?

Existing replay and curriculum methods rank examples by difficulty, temporal-difference error, regret, novelty, or fail-

ure. These scalar signals are useful, but they do not explicitly model the local response curve around a state or decide which perturbation radius should be trained once a state is selected. BGR instead estimates a state-conditioned recovery curve and trains near its success–failure transition. The empirical question is therefore not whether another priority score can win small paired comparisons; it is whether boundary-local radius selection adds anything beyond uniform replay, fixed radii, failure mining, and loss-priority proxies.

The contributions are:

- a replay interface based on replayable states, perturbation families, recovery curves, critical radii, and recovery AUC;
- a BGR algorithm that estimates success–failure boundaries and samples states and perturbation radii near them;
- active-estimation, synthetic, grid-margin, robot-suffix simulator, and OpenML supervised margin checks showing where boundary replay and coverage-aware boundary replay expand recovery margins, including a 30-seed grid ablation where state-priority replay with uniform radii loses the effect;
- negative standard-environment and OpenVLA/LIBERO audits separating recovery-margin measurement from robotics fine-tuning claims.

The claim is intentionally scoped. BGR is not a universal robustness guarantee, broad standard-environment result, or completed robotics fine-tuning method. It is a replay rule for settings with revisitable states, feasible perturbations, and trainable recovery curves; the OpenML, suffix, and OpenVLA/LIBERO studies test interface transfer and expose where learned-policy claims still need stronger evidence.

## 2 Related Work

BGR is closest to curriculum learning and level replay. Prioritized Level Replay samples levels by value-loss learning-potential proxies (Jiang, Grefenstette, and Rocktäschel 2021); PAIRED and ACCEL use regret and curriculum-frontier objectives (Dennis et al. 2020; Parker-Holder et al. 2022). The novelty claim is narrow and falsifiable: BGR changes the target from a level or transition score to an estimated curve crossing and the replay action from revisit-

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ing a hard state to choosing a radius near that crossing. If state-priority replay with uniform radii or loss-priority replay matches BGR, the boundary claim collapses to ordinary hard-example replay. The radius-level ablation is therefore a negative control: state priority is held fixed and only the radius rule changes. Drawing radii uniformly removes the grid-margin gain. BGR also differs from broad curriculum schedules (Narvekar et al. 2020): progress means recovery-margin expansion around each state, not global task difficulty.

Prioritized experience replay ranks transitions by temporal-difference error (Schaul et al. 2016), and active learning selects informative labels under a query budget (Settles 2009). BGR borrows the budgeted-query spirit but estimates a success-probability curve around a replay state, then trains at a state-conditioned radius. It is also distinct from adversarial-example training (Goodfellow, Shlens, and Szegedy 2015; Madry et al. 2018): BGR is not an adversarial-training objective and does not solve a worst-case inner maximization; it samples feasible scales near the observed boundary to expand local margin.

BGR also connects to robustness training and robotics generalization. Domain randomization broadens training distributions (Tobin et al. 2017), but fixed perturbation distributions can oversample states that are already easy or infeasible. BGR estimates each state’s success–failure crossing and trains near that radius. Our procedural experiments use resettable grid states in the style of lightweight grid benchmarks (Chevalier-Boisvert et al. 2023); the suffix study instantiates the interface for manipulation-style states related to LIBERO (Liu et al. 2023) and OpenVLA (Kim et al. 2024).

### 3 Problem Setting

Let  $\pi_\theta$  be a sequential decision policy and let  $Y(\tau) \in \{0, 1\}$  denote terminal success for rollout  $\tau$ . A replayable decision state  $x \in \mathcal{X}$  is any object from which evaluation or training can restart, such as an environment seed, simulator checkpoint, task prefix, or robot suffix state.

For each  $x$ , a domain-valid perturbation family applies  $\tilde{x} = T(x, \delta)$  with  $\delta \sim \mathcal{P}_x(\sigma)$ , where  $\sigma \geq 0$  is a perturbation magnitude. The perturbation family is part of the benchmark definition: grid perturbations preserve free cells and shortest-path reachability, OpenML feature perturbations preserve the supervised label by construction, suffix perturbations preserve teacher feasibility, and OpenVLA observation perturbations preserve the task instruction and simulator state while modifying the visual input stream. The recovery curve is

$$R_\pi(x, \sigma) = \Pr_{\delta, \tau}[Y(\tau) = 1 \mid \tau \sim \pi(\cdot \mid T(x, \delta))]. \quad (1)$$

For threshold  $\alpha$ , the critical radius is

$$r_{\pi, \alpha}(x) = \sup\{\sigma : R_\pi(x, \sigma) \geq \alpha R_\pi(x, 0)\}. \quad (2)$$

We evaluate robustness with recovery AUC,

$$\text{RAUC}_\pi(x) = \frac{1}{\sigma_{\max}} \int_0^{\sigma_{\max}} R_\pi(x, \sigma) d\sigma. \quad (3)$$

| Study          | Replay state           | Perturbation          | Witness/eval        |
|----------------|------------------------|-----------------------|---------------------|
| Synthetic      | latent recovery state  | scalar radius         | dense curve         |
| Grid-margin    | mid-path grid state    | valid radius shifts   | reachability        |
| Robot suffix   | demo suffix phase      | object/EEF offsets    | teacher feasibility |
| OpenML tabular | labeled feature vector | standardized L2 noise | label consistency   |
| OpenVLA audit  | LIBERO state           | object/visual changes | simulator success   |

Table 1: Instantiations of the BGR interface. The first three rows are training experiments; the OpenML row is a supervised margin check; the OpenVLA row is an audit of recovery curves and data plumbing, not a fine-tuning claim.

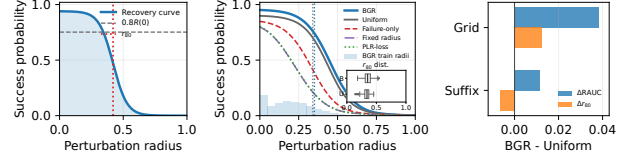


Figure 1: Geometric intuition for BGR, empirical trace, and metric cross-checks. Left: a recovery curve maps perturbation radius to success probability, and the critical radius is the success–failure crossing relative to clean success. Middle: a deterministic seed-0 grid-margin replay trace overlays final recovery curves for BGR, uniform, failure-only, fixed-radius, and PLR-loss replay with the BGR training-radius histogram and an inset showing BGR/uniform critical-radius distributions. Right: paired mean differences against uniform expose the metric tension: grid improves both RAUC and median  $r_{80}$ , while the suffix variant improves RAUC but reduces median  $r_{80}$ .

BGR depends on a feasibility witness  $F(x, \sigma)$  estimating whether perturbed states remain meaningful and solvable by a reference process. This is a real interface assumption, not free supervision or a learned success oracle: without it, boundary replay can waste updates on invalid perturbations or inflate margins with impossible states. Our witnesses are exact reachability checks, supervised label consistency, teacher/demo suffix models, simulator validity checks, or observed clean OpenVLA success. When no reliable witness is available, BGR should be treated as an audit tool or not applied. We only promote settings where the witness is exact or stress-tested. A 30-seed grid-margin witness diagnostic keeps exact-witness valid accepted samples at 1.0000; symmetric 10%/20% witness noise preserves valid-accept rates at 1.0000/0.9999 but lowers true-boundary recall to 0.9001/0.7980.

### 4 Boundary-Guided Replay

BGR maintains a buffer of replayable states. For each state, it probes perturbation radii, fits a monotone recovery curve, estimates a critical radius, and assigns replay priority to states whose boundaries are feasible, sharp, uncertain, and near a trainable target band. The method therefore separates three decisions: whether a state is valid enough to replay, which replay state is currently informative, and which perturbation radius lies near that state’s success–failure transition.

**Boundary-Guided Replay**

Initialize replay buffer  $\mathcal{B}$  with replayable states.

For each  $x \in \mathcal{B}$ , probe  $\sigma \in \{0, \sigma_{\min}, \sigma_{\text{mid}}, \sigma_{\max}\}$ , fit monotone  $\hat{R}(x, \sigma)$ , and estimate  $\hat{r}_\alpha(x)$ .

For each training iteration, sample  $x \sim q_{\mathcal{B}}(x)$  proportional to BGR priority, sample  $\sigma$  from a mixture centered at  $\hat{r}_\alpha(x)$ , apply  $\delta \sim \mathcal{P}_x(\sigma)$ , collect rollout or teacher recovery data, update  $\pi_\theta$ , and periodically refresh stale curve estimates.

*Method box: BGR training loop without domain-specific learner details.*

The priority for a record is a product of gates and soft utilities:

$$P(x) = G(x)U_{\text{boundary}}(x)U_{\text{sharp}}(x)^\lambda U_{\text{unc}}(x)^\beta U_{\text{stale}}(x)^\eta. \quad (4)$$

Here  $G(x)$  gates out states with low clean success or low witness feasibility,  $U_{\text{boundary}}$  prefers critical radii in a trainable band,  $U_{\text{sharp}}$  prefers informative transitions,  $U_{\text{unc}}$  refreshes uncertain curve estimates, and  $U_{\text{stale}}$  revisits old records as the policy changes. We mix this distribution with uniform replay to prevent starvation. For the radius-level decision, BGR samples from a mixture centered at  $\hat{r}_\alpha(x)$  with clean, easier, harder, and broad-radius coverage components; ablations test which part is responsible for the gains.

Two separations are important. First, BGR does not equate failure with usefulness: perturbations far beyond the current recovery radius can be valid but uninformative for margin expansion, while clean or near-clean states may already be saturated. Second, BGR separates replay-state priority from perturbation-radius choice. A state can be worth revisiting because its boundary is uncertain or sharp, but the training sample still needs to land near the recoverable transition rather than at an arbitrary radius. This distinction is what the radius-level ablation tests.

The following local calculation is a design rationale for the radius sampler and for the radius-level ablation, not a convergence result, global robustness theorem, or global margin-expansion guarantee.

**Local boundary intuition.** Fix a replay state  $x$ , current policy  $\pi$ , and critical radius  $r = r_{\pi, \alpha}(x)$ . Let  $\Delta_x(\sigma)$  denote the expected first-order change in held-out RAUC after one small update from a perturbed rollout at radius  $\sigma$ , holding the state distribution, learner linearization, and feasibility witness fixed. Assume this local gain depends only on distance to the current boundary,  $\Delta_x(\sigma) = k_x(|\sigma - r|)$ , for a nonincreasing kernel  $k_x$ . If a boundary sampler  $p_b(\sigma | x)$  induces a distance-to-boundary random variable  $D_b = |\sigma - r|$  satisfying  $\Pr[D_b > t] \leq \Pr[D_0 > t]$  for every  $t$  relative to a baseline sampler’s distance  $D_0$ , then

$$\mathbb{E}_{p_b}[k_x(|\sigma - r|)] \geq \mathbb{E}_{p_0}[k_x(|\sigma - r|)]. \quad (5)$$

The only mathematical assumption is first-order stochastic dominance over distance to the boundary plus monotonicity of the local gain kernel. By the quantile coupling for stochastic orders (Shaked and Shanthikumar 2007), there exists a coupling with  $D'_b \leq D'_0$  almost surely and with the same marginals as  $D_b, D_0$ ; monotonicity of  $k_x$  then gives  $\mathbb{E}[k_x(D'_b)] \geq \mathbb{E}[k_x(D'_0)]$ , which is the stated inequality. A

| Evidence tier  | Supports                             | Does not support                     |
|----------------|--------------------------------------|--------------------------------------|
| Synthetic      | sampler/estimator sanity             | independent benchmark win            |
| Grid-margin    | boundary-radius mechanism            | broad environment dominance          |
| Robot suffix   | manipulation-style interface         | final robotics fine-tuning claim     |
| OpenML tabular | pre-existing supervised margin check | standard-environment or robotics win |
| Gym/OpenVLA    | scope and audit checks               | promoted positive result             |

Table 2: Evidence contract used throughout the experiments. The central claim is mechanism-level margin expansion when recovery curves are directly trainable; standard-environment and learned-policy wins remain open requirements rather than hidden claims.

local margin-shift corollary makes the implication concrete: if one small update at radius  $\sigma$  shifts the current critical radius right by at least  $c k_x(|\sigma - r|)$  and never shifts it left, then the expected one-step critical-radius gain under  $p_b$  is at least that under  $p_0$ , and any local RAUC lower bound proportional to this shift inherits the same ordering. This is still only a one-step, fixed-state statement. BGR uses boundary-centered sampling as a default but retains clean, easier, harder, broad-radius, and uniform components because real learners can violate the local model through saturation, noisy labels, witness errors, or unstable optimization.

**Finite-grid estimator guarantee.** For  $N$  states,  $K$  radii, and  $m$  Bernoulli rollouts per state-radius pair, Hoeffding plus a union bound gives  $\max_{x,j} |\bar{R}(x, \sigma_j) - R(x, \sigma_j)| \leq \epsilon = \sqrt{\log(2NK/\delta)/(2m)}$  with probability  $1 - \delta$ . If the true crossings have local slope at least  $\gamma$  and grid spacing is at most  $\Delta$ , then every estimated critical radius is within  $\Delta + (1 + \alpha)\epsilon/\gamma$ ; achieving error  $\eta > \Delta$  scales as  $m = O(\log(NK/\delta)/(\gamma^2(\eta - \Delta)^2))$ . This certifies boundary localization, not learner improvement.

## 5 Experiments

All experiments are driven by artifact YAML configs and scripts. The anonymous artifact records per-seed CSVs, aggregation code, paired exact sign tests, Slurm launch commands, run ledger files, compute/GPU environment snapshots, and an MIT license; the paper tables are generated from these artifacts rather than hand-edited results.

Table 2 summarizes the claim boundary before the results. The main claim is supported by controlled synthetic and procedural grid-margin experiments where the recovery-margin object is directly trainable; the grid result is replicated on held-out seeds and pooled over 60 paired comparisons. Fixed OpenML diabetes and blood-transfusion checks add pre-existing supervised-dataset evidence for the same radius-selection mechanism. The robot-suffix simulator is a manipulation-style extension: a coverage-aware BGR variant improves final object RAUC, clean success, transfer RAUC, and AULC over uniform suffix replay across original and held-out sweeps and in four stress regimes, while still trailing uniform on median critical radius. Gym-style standard-environment probes and OpenVLA/LIBERO results are scope checks and learned-policy audits, not promoted positive claims.

## 5.1 Protocol and Metrics

We report clean success, recovery AUC (RAUC), median  $r_{80}$ , and area under the RAUC learning curve (AULC), evaluated on held-out radius grids. RAUC and AULC are author-defined summaries, so we report clean success and median  $r_{80}$  separately and discuss disagreements directly. Synthetic training, estimator validation, packaged grid confirmations, OpenML checks, and suffix confirmations use 30 paired seeds unless noted; grid learning curves use 15. We treat exact paired sign tests as consistency checks over shared seeds, not as substitutes for effect size: a 30/0 sign pattern can certify direction while the absolute gain remains small. The synthetic and suffix gains are therefore modest positive effects, while the grid-margin mechanism benchmark shows the largest practically visible gap. Five-seed exploratory variants and small pre-promotion standard-environment probes are reported only as scope diagnostics.

Before promoting a new result as positive evidence, we require it to use a fixed protocol, beat uniform replay on final RAUC with a visible effect size, beat the relevant fixed-radius and priority baselines, avoid contradictory complementary metrics or report them as limitations, and either replicate on held-out seeds or pass a preregistered stress sweep. Uniform, fixed-radius, failure-only, PLR-loss/loss-priority, and clean-only baselines are matched to the replay decision being tested; paired comparisons share seeds, roll-out budgets, evaluation grids, and validity filters.

## 5.2 Active Boundary Estimation

BGR is useful only if recovery boundaries can be estimated cheaply. We therefore validate the training-time estimator separately: for each synthetic state, a dense 201-point reference curve is used only for evaluation, and 17-probe uniform radii are compared with active BGR probing. Table 5 shows that active BGR finds the boundary within  $\pm 0.08$  radius units more often than uniform probing (0.670 vs. 0.595) and lowers mean  $r_{80}$  error (0.081 vs. 0.105). The paired consistency is reported in Table 4; the practical point is the fixed-budget effect size, not the p-value.

## 5.3 Synthetic Recovery-Margin Benchmark

The synthetic benchmark creates replayable states with latent feasible radii and sigmoid recovery curves, deliberately exposing the mechanism BGR targets rather than independent benchmark evidence. In the 30-seed run, BGR gives a modest final recovery-AUC gain over uniform replay (0.3716 vs. 0.3633; mean difference 0.0084), a larger learning-curve gain (AULC 0.3149 vs. 0.2984), and similar clean success (0.8899 vs. 0.8838). Fixed-radius, failure-only, and PLR-loss proxy baselines underperform substantially in this controlled setting.

## 5.4 Procedural Grid-Margin Study

We instantiate replayable states in procedural grid environments with resettable seeds, mid-path states, shortest-path reachability witnesses, and policy-specific recovery margins. In a 30-seed full-baseline run, BGR gives the largest absolute gap in the paper: final RAUC 0.4344 vs. 0.3963,

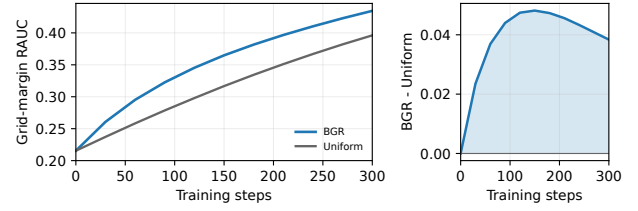


Figure 2: Grid-margin RAUC learning curve from the stored 15-seed run. Shaded bands show standard error. The right panel plots the paired mean gap, making the effect size visible without relying on significance notation.

RAUC learning-curve area 0.3526 vs. 0.3132, median  $r_{80}$  0.3447 vs. 0.3321, and clean success 0.9455 vs. 0.8939. A held-out seeds 30–59 replication gives 0.4340 vs. 0.3967 RAUC; pooled original and held-out sweeps give 0.4342 vs. 0.3965 RAUC with 60/0 paired wins. Failure-only replay (0.2909), PLR-loss proxy replay (0.2109), and fixed-radius training (0.2097) substantially underperform BGR (0.4344).

Figure 2 shows stored 15-seed learning curves: BGR separates early and remains ahead through steps 30–300. The endpoint gap is about 0.038 RAUC, the paper’s clearest practical effect but still scoped mechanism evidence. Thirty-seed sensitivities keep the pattern across target margins 0.26–0.54 (RAUC 0.416–0.443 vs. uniform 0.396), obstacle regimes (0.434–0.435 vs. 0.396), and geometry stresses (0.362–0.457 vs. 0.323–0.430). The learning-rate table preserves the high-rate final-RAUC caveat, and the reported 0.38 margin is not the best tuned value.

A learning-rate sweep is a scope diagnostic: BGR wins final RAUC at 0.015 and 0.030 (0.382 vs. 0.320; 0.434 vs. 0.396) and AULC at all three rates, but high learning rate flips final RAUC to uniform (0.485 vs. 0.491). Thus neither failure mining nor static perturbation scales recover the same margin-expansion behavior, while aggressive optimization can saturate uniform final RAUC.

| Method        | Clean       | RAUC        | Median $r_{80}$ | AULC        |
|---------------|-------------|-------------|-----------------|-------------|
| BGR           | 0.945±0.000 | 0.434±0.000 | 0.345±0.001     | 0.353±0.000 |
| Uniform       | 0.894±0.000 | 0.396±0.000 | 0.332±0.001     | 0.313±0.000 |
| Failure-only  | 0.845±0.000 | 0.291±0.000 | 0.231±0.001     | 0.254±0.000 |
| Loss-priority | 0.797±0.001 | 0.211±0.000 | 0.137±0.001     | 0.213±0.000 |
| Fixed radius  | 0.794±0.001 | 0.210±0.000 | 0.136±0.001     | 0.213±0.000 |

Table 6: Thirty-seed procedural grid-margin full-baseline comparison. Boundary-centered replay is the only method that both preserves clean success and expands recovery margins.

| Method                         | Clean       | RAUC        | Median $r_{80}$ | AULC        |
|--------------------------------|-------------|-------------|-----------------|-------------|
| BGR                            | 0.945±0.000 | 0.434±0.000 | 0.345±0.001     | 0.353±0.000 |
| No uncertainty                 | 0.946±0.000 | 0.435±0.000 | 0.344±0.001     | 0.353±0.000 |
| No sharpness                   | 0.946±0.000 | 0.435±0.000 | 0.344±0.001     | 0.353±0.000 |
| BGR state + uniform radius     | 0.891±0.000 | 0.382±0.000 | 0.322±0.001     | 0.305±0.000 |
| Uniform state + uniform radius | 0.894±0.000 | 0.396±0.000 | 0.332±0.001     | 0.313±0.000 |

Table 7: Grid-margin ablations over 30 seeds. Boundary-centered radius sampling is the important ingredient in this benchmark: replacing it with uniform radii drops RAUC below uniform replay, while removing uncertainty or sharpness weights has little effect.

Table 7 isolates the replay decisions inside BGR. The uncertainty and sharpness factors are not decisive in this benchmark, but the radius-level boundary rule is: keeping BGR

| Benchmark       | Method        | Clean       | RAUC        | Median $r_{80}$ | AULC        |
|-----------------|---------------|-------------|-------------|-----------------|-------------|
| Synthetic       | BGR           | 0.890±0.001 | 0.372±0.001 | 0.286±0.002     | 0.315±0.001 |
| Synthetic       | Uniform       | 0.884±0.001 | 0.363±0.001 | 0.282±0.002     | 0.298±0.001 |
| Synthetic       | Fixed         | 0.821±0.001 | 0.230±0.001 | 0.131±0.001     | 0.229±0.001 |
| Synthetic       | Failure       | 0.817±0.001 | 0.233±0.001 | 0.135±0.001     | 0.231±0.001 |
| Synthetic       | Loss-priority | 0.819±0.001 | 0.229±0.001 | 0.134±0.001     | 0.229±0.001 |
| GridMargin      | BGR           | 0.945±0.000 | 0.434±0.000 | 0.345±0.001     | 0.353±0.000 |
| GridMargin      | Uniform       | 0.894±0.000 | 0.396±0.000 | 0.332±0.001     | 0.313±0.000 |
| OpenML diabetes | BGR           | —           | 0.706±0.006 | —               | —           |
| OpenML diabetes | Uniform       | —           | 0.669±0.010 | —               | —           |
| OpenML diabetes | Fixed-radius  | —           | 0.676±0.007 | —               | —           |
| OpenML blood    | BGR           | —           | 0.762±0.002 | —               | —           |
| OpenML blood    | Uniform       | —           | 0.666±0.011 | —               | —           |
| OpenML blood    | Fixed-radius  | —           | 0.692±0.012 | —               | —           |
| RobotSuffix     | BGR-Coverage  | 0.864±0.001 | 0.497±0.001 | 0.498±0.001     | 0.382±0.001 |
| RobotSuffix     | Uniform       | 0.836±0.001 | 0.485±0.001 | 0.505±0.001     | 0.371±0.001 |

Table 3: Thirty-seed result summary from artifact CSVs. OpenML rows are supervised margin-replay checks, so only RAUC applies; held-out replications for grid, OpenML, and suffix are reported in text. BGR is strongest in the procedural grid-margin and OpenML blood-transfusion checks. BGR-Suffix improves clean success, final object RAUC, transfer RAUC, and AULC, but uniform keeps the higher median critical radius.

state priority while sampling perturbation radii uniformly reduces RAUC from 0.434 to 0.382 and AULC from 0.353 to 0.305, even below uniform replay’s 0.396 RAUC. A held-out seeds 30–59 replication gives 0.434 vs. 0.382 RAUC and 0.352 vs. 0.305 AULC for BGR over the uniform-radius ablation, again with 30/0 paired wins; uniform-radius remains below uniform replay (0.382 vs. 0.397 RAUC). This is the strongest novelty evidence in the paper: the gain is not explained by hard-state prioritization alone, because state priority is held fixed and only the radius rule changes.

## 5.5 Pre-existing OpenML Margin Check

To test whether the radius-selection mechanism appears outside authored recovery environments, we ran fixed supervised margin-replay checks on OpenML datasets. A four-seed scout over ionosphere, sonar, diabetes, and spam-base selected only OpenML diabetes at target radius 2.0; no target retuning or dataset expansion was used in the follow-up. The 30-seed fixed run gives BGR 0.7062 versus 0.6689 uniform (gap +0.0373; W/L/T=24/6/0) and 0.6759 fixed-radius (gap +0.0303; W/L/T=19/11/0). A held-out seeds 30–59 replication gives 0.7056 versus 0.6673 uniform (gap +0.0383; W/L/T=23/7/0) and 0.6640 fixed-radius (gap +0.0416; W/L/T=24/6/0); pooled over 60 seeds, BGR is 0.7059 versus 0.6681 uniform and 0.6699 fixed-radius.

We then ran a fixed external numeric OpenML validation suite at the same target radius over banknote-authentication, blood-transfusion-service-center, climate-model-simulation-crashes, kc1, mozilla4, pc1, phoneme, and wdbc. The suite is mixed: macro means are BGR 0.8055, uniform 0.7939, and fixed-radius 0.7864, with several negative or below-gate datasets. Blood-transfusion clears the fixed gate: BGR reaches 0.7625 versus 0.6657

uniform (gap +0.0968; W/L/T=30/0/0) and 0.6920 fixed-radius (gap +0.0705; W/L/T=27/1/2). Held-out seeds 30–59 replicate at 0.7595 versus 0.6846 uniform (gap +0.0749; W/L/T=25/5/0) and 0.7133 fixed-radius (gap +0.0462; W/L/T=25/5/0); pooled over 60 seeds, BGR is 0.7610 versus 0.6751 uniform and 0.7026 fixed-radius. Phoneme also replicates: the original row gives 0.7228 versus 0.6896 uniform and 0.6704 fixed-radius (gap +0.0524; W/L/T=23/7/0), while held-out seeds give 0.7124 versus 0.6758 uniform (gap +0.0366; W/L/T=21/9/0) and 0.6792 fixed-radius (gap +0.0332; W/L/T=25/5/0); pooled over 60 seeds, BGR is 0.7176 versus 0.6827 uniform and 0.6748 fixed-radius. These are pre-existing supervised-dataset checks for boundary-radius replay, not standard-environment recovery or robotics results.

## 5.6 Robot Suffix Study

BGR-Suffix treats a robot demonstration suffix as a replayable state in a lightweight language-conditioned manipulation simulator with object-pose and end-effector perturbations, teacher feasibility, and clean-demo regularization.

The first 15-seed run showed that boundary-heavy BGR-Suffix undercovered high-radius recovery. We therefore treat the coverage-aware variant as a separate manipulation-style support result, not evidence that boundary-only replay is robust: it adds broad-radius and uniform-state coverage and is promoted only after 30-seed confirmation, held-out replication, and stress sweeps. In the 30-seed run, BGR-Coverage raises final object RAUC over uniform by a small but consistent amount (0.4969 vs. 0.4854), with clean success 0.8644 vs. 0.8364, transfer RAUC 0.3143 vs. 0.3083, and AULC 0.3825 vs. 0.3709. A held-out seeds 30–59 replication gives 0.4972 vs. 0.4859 RAUC. Pooling origi-

| Comparison                      | Metric        | $n$ | Mean diff. | 95% CI           | W/L/T   | $p$        |
|---------------------------------|---------------|-----|------------|------------------|---------|------------|
| Synthetic vs uniform            | final RAUC    | 30  | 0.0083     | [0.0059, 0.0107] | 29/1/0  | $< 0.0001$ |
| Grid vs uniform                 | final RAUC    | 30  | 0.0381     | [0.0372, 0.0390] | 30/0/0  | $< 0.0001$ |
| OpenML diabetes vs uniform      | final RAUC    | 30  | 0.0373     | [0.0152, 0.0594] | 24/6/0  | 0.0014     |
| OpenML diabetes vs fixed-radius | final RAUC    | 30  | 0.0303     | [0.0081, 0.0525] | 19/11/0 | 0.2005     |
| OpenML blood vs uniform         | final RAUC    | 30  | 0.0968     | [0.0739, 0.1198] | 30/0/0  | $< 0.0001$ |
| OpenML blood vs fixed-radius    | final RAUC    | 30  | 0.0705     | [0.0474, 0.0937] | 27/1/2  | $< 0.0001$ |
| Suffix RAUC vs clean-only       | final RAUC    | 30  | 0.2338     | [0.2327, 0.2350] | 30/0/0  | $< 0.0001$ |
| Suffix RAUC vs fixed            | final RAUC    | 30  | 0.3386     | [0.3373, 0.3399] | 30/0/0  | $< 0.0001$ |
| Suffix RAUC vs failure-only     | final RAUC    | 30  | 0.0741     | [0.0731, 0.0750] | 30/0/0  | $< 0.0001$ |
| Suffix RAUC vs loss-priority    | final RAUC    | 30  | 0.3275     | [0.3262, 0.3288] | 30/0/0  | $< 0.0001$ |
| Suffix RAUC vs uniform          | final RAUC    | 30  | 0.0116     | [0.0109, 0.0122] | 30/0/0  | $< 0.0001$ |
| Suffix transfer vs uniform      | transfer RAUC | 30  | 0.0061     | [0.0054, 0.0068] | 30/0/0  | $< 0.0001$ |
| Suffix AULC vs uniform          | RAUC AULC     | 30  | 0.0115     | [0.0108, 0.0122] | 30/0/0  | $< 0.0001$ |
| Estimator 30-seed               | boundary hit  | 30  | 0.0751     | [0.0646, 0.0857] | 30/0/0  | $< 0.0001$ |

Table 4: Selected paired effect sizes. Mean differences and 95% confidence intervals are the primary quantities; W/L/T and exact sign-test  $p$  values report paired-seed direction consistency. All listed comparisons use 30 paired seeds.

| Estimator  | $r_{80}$ MAE      | RAUC MAE          | Hit rate          |
|------------|-------------------|-------------------|-------------------|
| Active BGR | 0.081 $\pm$ 0.001 | 0.065 $\pm$ 0.000 | 0.670 $\pm$ 0.004 |
| Uniform    | 0.105 $\pm$ 0.001 | 0.066 $\pm$ 0.000 | 0.595 $\pm$ 0.004 |

Table 5: Probe-efficiency validation on synthetic recovery curves. Both methods use 17 Bernoulli probes per state and are evaluated against dense 201-point curves. Active BGR improves boundary-hit rate over uniform probing at the same rollout budget, while keeping critical-radius and RAUC error comparable.

nal and held-out suffix sweeps gives object RAUC 0.4971 vs. 0.4856, clean 0.8642 vs. 0.8367, transfer RAUC 0.3150 vs. 0.3089, and AULC 0.3829 vs. 0.3717. A 30-seed strategy ablation gives 0.4969 vs. 0.4587 and 0.4850 final object RAUC for coverage-aware, boundary-heavy, and hard-radius BGR; the hard-radius variant leads transfer RAUC and AULC (0.3252, 0.3928), but only coverage-aware replay improves final object RAUC over uniform with 30/0 wins.

A four-condition 30-seed suffix stress sweep varies teacher quality, clutter, feasible radii, and boundary sharpness. BGR-Coverage keeps 30/0 paired wins over uniform on clean success, final object RAUC, end-effector transfer RAUC, and AULC in every stress case, with object RAUC 0.3806–0.4971 vs. uniform 0.3755–0.4853 and transfer RAUC 0.2267–0.3142 vs. 0.2205–0.3081. Clean-only has a negligible clean-success edge (0.8646 vs. 0.8644), and uniform remains higher on median  $r_{80}$  pooled (0.5048 vs. 0.4985), in the original sweep (0.5047 vs. 0.4982), and in the stress sweep. We therefore treat suffix results as manipulation-style evidence, not final robotics evidence.

We audited the real-simulator path as premise validation. A GPU/EGL LIBERO-Goal probe completed 75/75 valid radius rows, and fixed-policy OpenVLA audits show clean replay success of 1.0, recovery drops under occlusion, blur, and shift, and BGR-boundary selection in the observed

counterfactual-failure band  $[0.25, 0.75]$  slightly more often than random-balanced selection (0.625 vs. 0.575).

The OpenVLA-OFT bridge exports 11,776 teacher-action rows; action-label/TFDS plumbing validates 2,048-transition matched BGR/random exports with 7D actions and 8D state, stock loader ingestion, and matched 10-step LoRA checkpoint smokes. The adaptation audits remain negative: Identity is 99/100 for BGR, matched random, and official in the full-goal audit; visual perturbation gives BGR 367/400, official 367/400, and random 368/400; Image augmentation gives BGR/random 368/400 vs. official 367/400 but worse BGR identity; low-LR continuation gives BGR 366/400 vs. official 367/400 and random 370/400; proximal anchor ties random at 368/400 with BGR identity 98/100 vs. official 99/100; Perturb-only anchoring preserves identity at 99/100 and raises BGR to 371/400, but official is 367/400 and random is 372/400, so BGR still fails the fixed +10/400 and 0.02 gate. Across these runs there is no stable official improvement or learned-policy improvement over both comparators.

## 6 Limitations

BGR requires reset or replay access, a meaningful perturbation family, a feasibility witness, and extra rollouts for boundary estimation. It targets an observed boundary, not a formal robustness certificate. The clearest evidence is controlled procedural margin expansion; the synthetic and grid-margin benchmarks are constructed to expose recovery curves, so they establish mechanism rather than broad benchmark dominance. The replicated OpenML margin checks reduce the pre-existing-dataset concern but are supervised margin checks, not standard-environment recovery benchmarks. The paper still does not claim a clear win on standard environments or learned-policy robotics fine-tuning.

[illegible]

Table 8: Standard-environment scope audits (limitations, not positive claims).

Table 8 keeps the scope explicit. FrozenLake has a small BGR RAUC edge but loses paired-sign and complementary-metric checks; MiniGrid, PointMaze, FetchReach, Reacher, bsuite, and MinAtar fail against stronger baselines, radius checks, or paired wins; and InvertedPendulum ties all methods. LunarLander is a near miss: BGR-Coverage has the best mean RAUC (0.7500 vs. 0.6222 uniform) but wins only 2/4 seeds and has lower median  $r_{80}$ . bsuite Catch shows scale-up fragility (0.8446 vs. 0.8782 uniform), bsuite Cartpole trails TD-loss replay, MinAtar Breakout ties uniform and failure-only at 0.8896 RAUC with saturated  $r_{80} = 5.0000$ , and MinAtar Asterix trails failure-only (0.8406 vs. 0.8625 RAUC). InvertedDoublePendulum-v5 gives a tiny BGR RAUC edge (0.0833 vs. 0.0035 uniform) but collapses clean success to 0.2500 for BGR and 0.0000 for BGR-Coverage. FourRooms radius-10, HandReach-v3, and highway-fast-v0 follow-ups close simple rescue paths with saturated or invalid recovery curves. Earlier sklearn digits, breast-cancer, wine, and non-diabetes OpenML scouts remain internal diagnostics rather than manuscript evidence.

The suffix simulator is positive on final RAUC, clean success, transfer RAUC, and AULC, including stress regimes, but gains are small and uniform retains the median-critical-radius edge. RAUC and AULC are useful for measuring curve expansion, but they are author-defined integrals over chosen radius grids; we report median  $r_{80}$  and clean success separately instead of treating RAUC alone as decisive. OpenVLA is a learned-policy audit rather than a robotics training claim: it validates recovery-curve measurement and matched action/TFDS construction, while showing that a full robotics claim still requires stable optimization, clean-data scale, and LIBERO gains over both matched random and the official checkpoint.

## 7 Conclusion

BGR converts recovery-margin measurement into a replay curriculum. With reset access, feasible perturbations, and directly trainable recovery curves, boundary replay improves margin expansion over uniform, fixed-radius, failure-only, and loss-proxy replay; hard-state priority alone loses the grid-margin effect when radii are drawn uniformly. Synthetic and grid-margin studies establish the mechanism; suffix results show scoped manipulation-style transfer. The current learned-policy evidence remains an audit, not the final robotics result.

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# Reproducibility Checklist

This checklist is filled for the current BGR submission package. The main paper is anonymous; commands, configs, per-seed CSVs, and run ledgers are in the anonymous artifact's `README.md` and `results/README.md`.

## 1. General Paper Structure

- 1.1. Includes a conceptual outline and/or pseudocode description of AI methods introduced (yes/partial/no/NA)  
[yes. The method section defines replayable states and recovery curves. It also states critical radii, BGR priority, and a training-loop method box.](#)
- 1.2. Clearly delineates statements that are opinions, hypothesis, and speculation from objective facts and results (yes/no)  
[yes. The paper separates controlled margin-expansion claims, robot-suffix simulator results, OpenVLA fixed-policy audits, OpenVLA-OFT adaptation diagnostics, and tabular grid-policy setup failures.](#)
- 1.3. Provides well-marked pedagogical references for less-familiar readers to gain background necessary to replicate the paper (yes/no)  
[yes. The related-work section cites curriculum learning, PLR, PAIRED/ACCEL, domain randomization, MiniGrid, LIBERO, and OpenVLA.](#)

## 2. Theoretical Contributions

- 2.1. Does this paper make theoretical contributions? (yes/no)  
[yes. The paper introduces a recovery-margin formalism and states a local margin-update proposition for boundary-centered radius sampling.](#)

If yes, please address the following points:

- 2.2. All assumptions and restrictions are stated clearly and formally (yes/partial/no)  
[yes. The local boundary-intuition calculation states the local update-kernel and stochastic-dominance assumptions; the limitations section states replay-access and perturbation-family restrictions.](#)
- 2.3. All novel claims are stated formally (e.g., in theorem statements) (yes/partial/no)  
[yes. The local theoretical intuition is stated as a design rationale. Empirical claims are stated with measured metrics and paired-seed tests.](#)
- 2.4. Proofs of all novel claims are included (yes/partial/no)  
[yes. The local boundary-intuition calculation includes a proof sketch based on monotone coupling and monotonicity of the update kernel.](#)
- 2.5. Proof sketches or intuitions are given for complex and/or novel results (yes/partial/no)  
[yes. The method section gives the local margin-update intuition and the proposition proof sketch.](#)
- 2.6. Appropriate citations to theoretical tools used are given (yes/partial/no)  
[yes. The proof cites a standard stochastic-orders reference for the quantile coupling used in the local boundary-intuition calculation.](#)
- 2.7. All theoretical claims are demonstrated empirically to hold (yes/partial/no/NA)  
[partial. The proposition is a local model; the grid ablation empirically tests its main implication that boundary-centered radius sampling matters.](#)
- 2.8. All experimental code used to eliminate or disprove claims is included (yes/no/NA)  
[yes. Ablation scripts, configs, per-seed CSVs, and claim-checking scripts are included.](#)

## 3. Dataset Usage

- 3.1. Does this paper rely on one or more datasets? (yes/no)  
[partial. Main BGR training results use procedural synthetic/grid/suffix generators. The LIBERO/OpenVLA recovery audit uses existing benchmark task definitions, init states, and fixed-policy rollout artifacts to validate learned-policy recovery-curve measurement.](#)

If yes, please address the following points:

- 3.2. A motivation is given for why the experiments are conducted on the selected datasets (yes/partial/no/NA)  
[yes. The paper motivates synthetic recovery curves, procedural grid-margin replay states, robot suffix diagnostics, and LIBERO/OpenVLA as a realistic manipulation-policy target.](#)
- 3.3. All novel datasets introduced in this paper are included in a data appendix (yes/partial/no/NA)  
[yes. The experiments are generated by artifact-provided scripts/configs; per-seed result CSVs are included under `results/`.](#)
- 3.4. All novel datasets introduced in this paper will be made publicly available upon publication of the paper with a license that allows free usage for research purposes (yes/partial/no/NA)  
[yes. The generated procedural data and code are included in the anonymous artifact under an MIT license.](#)
- 3.5. All datasets drawn from the existing literature (potentially including authors' own previously published work) are accompanied by appropriate citations (yes/no/NA)  
[yes. LIBERO and OpenVLA are cited; MiniGrid is cited for grid-style context.](#)



- 3.6. All datasets drawn from the existing literature (potentially including authors' own previously published work) are publicly available (yes/partial/no/NA)  
yes for LIBERO task definitions, benchmark assets, and OpenVLA model code/weights. The anonymous artifact contains compact derived recovery summaries; raw fixed-policy rollout logs are retained for audit but are not required for the claimed quantitative results.
- 3.7. All datasets that are not publicly available are described in detail, with explanation why publicly available alternatives are not scientifically satisfying (yes/partial/no/NA)  
NA. No non-public dataset is used for claimed quantitative results.

## 4. Computational Experiments

- 4.1. Does this paper include computational experiments? (yes/no)

yes.

If yes, please address the following points:

- 4.2. This paper states the number and range of values tried per (hyper-) parameter during development of the paper, along with the criterion used for selecting the final parameter setting (yes/partial/no/NA)  
partial. Final configs and ablation variants are included in the anonymous artifact; the full search history is not exhaustively enumerated in the main paper.
- 4.3. Any code required for pre-processing data is included in the appendix (yes/partial/no)  
yes. Procedural generation, LIBERO probing, OpenVLA recovery summarization, aggregation, and figure scripts are included.
- 4.4. All source code required for conducting and analyzing the experiments is included in a code appendix (yes/partial/no)  
yes. Experiment scripts, configs, tests, aggregation, and result CSVs are included in the anonymous artifact.
- 4.5. All source code required for conducting and analyzing the experiments will be made publicly available upon publication of the paper with a license that allows free usage for research purposes (yes/partial/no)  
yes. The anonymous artifact includes an MIT license, and the code plus generated procedural data will be made publicly available upon publication.
- 4.6. All source code implementing new methods have comments detailing the implementation, with references to the paper where each step comes from (yes/partial/no)  
partial. Code is modular and tested; comments are intentionally sparse and do not yet cross-reference paper sections.
- 4.7. If an algorithm depends on randomness, then the method used for setting seeds is described in a way sufficient to allow replication of results (yes/partial/no/NA)  
yes. Every config lists seeds; scripts pass seeds to NumPy generators and result CSVs include per-seed rows.
- 4.8. This paper specifies the computing infrastructure used for running experiments (hardware and software), including GPU/CPU models; amount of memory; operating system; names and versions of relevant software libraries and frameworks (yes/partial/no)  
yes. The run ledger records Slurm partitions, CPU/memory/time requests, hostnames for logged runs, environment constraints, and JSON environment snapshots for compute and GPU jobs.
- 4.9. This paper formally describes evaluation metrics used and explains the motivation for choosing these metrics (yes/partial/no)  
yes. Recovery AUC, critical radius, clean success, transfer RAUC, and AULC are defined or described.
- 4.10. This paper states the number of algorithm runs used to compute each reported result (yes/no)  
yes. Runs are reported in the paper and run ledger: synthetic training, estimator validation, grid-margin full-baseline, grid ablation, grid sensitivity confirmations, and robot-suffix coverage comparisons use 30 paired seeds; grid learning-curve diagnostics use 15 seed pairs. Smaller three- or five-seed exploratory diagnostics are labeled as diagnostics in the run ledger rather than used as main claims.
- 4.11. Analysis of experiments goes beyond single-dimensional summaries of performance (e.g., average; median) to include measures of variation, confidence, or other distributional information (yes/no)  
yes. Aggregated tables include standard errors; experiments report median critical radius and AULC in addition to final means.
- 4.12. The significance of any improvement or decrease in performance is judged using appropriate statistical tests (e.g., Wilcoxon signed-rank) (yes/partial/no)  
yes. The anonymous artifact includes paired exact sign tests for the reported paired comparisons, including the 30-seed suffix coverage confirmation, grid-margin target-sensitivity, grid-margin obstacle-regime sensitivity, grid-margin geometry-stress sensitivity, and grid-margin learning-curve sweeps. The paper separates these main comparisons from smaller exploratory evidence.
- 4.13. This paper lists all final (hyper-)parameters used for each model/algorithm in the paper's experiments (yes/partial/no/NA)  
yes. Final hyperparameters are in the artifact YAML configs under `configs/`.